ENGINEERING MATHEMATICS-II

UNIT 1: INTEGRAL CALCULUS

Class 12: moment of inertia

PHYSICAL IMPORTANCE OF MOMENTS OF INERTIA

Physical importance of moment of inertia: The moment of inertia is very important in our daily life. To increase the moment inertia of the wheels of scooter, motor cycle, bicycle, riksha, tanga and bull cart etc. most of the material i.e., is on the circumference which is connected by spokes with the axis. By riding on a bicycle and then leaving the pedal it can move for some time i. e., upto some distance due to more moment of inertia, which means the wheels keeps on moving. The use of moment of inertia is very significant in automobile areas. Automobile engines which generate rotational energy, has a flywheel which is a wheel with high moment of inertia. When the value of torque increases or decreases which rotates the shaft, then the flywheel due to very high moment of inertia moves with a constant speed, due to which we are saved from high shocks. The motor toys of children also have flywheel. It

MOMENTS OF INERTIA FOR A FUNCTION OF TWO INDEPENDENT VARIALBES

Suppose that $\rho(x,y)$ is a continuous density function on a lamina R.

Then the *moments of inertia* are

$$I_x = \iint_R \rho(x,y)\, y^2 \, dy\, dx$$

$$I_y = \iint_R \rho(x,y)\, x^2 \, dy\, dx$$

EXAMPLE 1: MOMENTS OF INERTIA FOR A FUNCTION OF TWO INDEPENDENT VARIALBES

Use the triangular region R with vertices $(0,0)\,,(2,2)\,,$ and $(2,0)$ and with density $\rho\,(x,y) = xy$

Using the expressions established above for the moments of inertia, we have

$$I_x = \iint\limits_R y^2 \rho\,(x,y)\,dA = \int\limits_{x=0}^{x=2} \int\limits_{y=0}^{y=x} xy^3 dy\,dx = \frac{8}{3},$$

$$I_y = \iint\limits_R x^2 \rho\,(x,y)\,dA = \int\limits_{x=0}^{x=2} \int\limits_{y=0}^{y=x} x^3 y\,dy\,dx = \frac{16}{3},$$

$$\begin{aligned} I_0 &= \iint\limits_R \left(x^2+y^2\right)\rho(x,y)dA = \int\limits_0^2 \int\limits_0^x \left(x^2+y^2\right) xy\,dy\,dx \\ &= I_x + I_y = 8. \end{aligned}$$

MOMENTS OF INERTIAL FOR A FUNCTION OF THREE INDEPENDENT VARIALBES ABOUT THE YZ-PLANE,XZ-PLANE AND XY-PLANE

The moments of inertia of a solid about the coordinate planes Oxy, Oxz, Oyz are given by

$$I_{xy} = \iiint\limits_U z^2 \rho(x, y, z)\, dxdydz, \quad I_{yz} = \iiint\limits_U x^2 \rho(x, y, z)\, dxdydz,$$

$$I_{xz} = \iiint\limits_U y^2 \rho(x, y, z)\, dxdydz,$$

MOMENTS OF INERTIAL FOR A FUNCTION OF THREE INDEPENDENT VARIALBES ABOUT THE YZ-PLANE,XZ-PLANE AND XY-PLANE

moments of inertia of a solid about the coordinate axes Ox, Oy, Oz are expressed by the

$$I_x = \iiint\limits_U \left(y^2 + z^2\right) \cdot \rho\left(x, y, z\right) dxdydz,\ \ I_y = \iiint\limits_U \left(x^2 + z^2\right) \cdot \rho\left(x, y, z\right) dxdydz,$$

$$I_z = \iiint\limits_U \left(x^2 + y^2\right) \cdot \rho\left(x, y, z\right) dxdydz.$$

As it can be seen, the following properties are valid:

$$I_x = I_{xy} + I_{xz},\ \ I_y = I_{xy} + I_{yz},\ \ I_z = I_{xz} + I_{yz}.$$

The moment of inertia about the origin is

$$I_0 = \iiint\limits_U \left(x^2 + y^2 + z^2\right) \cdot \rho\left(x, y, z\right) dxdydz.$$

EXAMPLE 1: MOMENTS OF INERTIA FOR A FUNCTION OF THREE INDEPENDENT VARIALBES

Suppose that Q is a solid region and is bounded by x+2y=3z=6 and the coordinate planes with density function

Find the moments of inertia of the tetrahedron about the x-axis, y axis and z=axis.

$$I_x = \iiint\limits_U \left(y^2 + z^2\right) \cdot \rho\left(x, y, z\right) dxdydz, \;\; I_y = \iiint\limits_U \left(x^2 + z^2\right) \cdot \rho\left(x, y, z\right) dxdydz,$$

$$I_z = \iiint\limits_U \left(x^2 + y^2\right) \cdot \rho\left(x, y, z\right) dxdydz.$$

$$I_x = \iiint_Q \left(y^2+z^2\right) x^2yz\,dV = \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} \left(y^2+z^2\right) x^2yz\,dz\,dy\,dx = \frac{117}{35} \approx 3.343,$$

$$I_y = \iiint_Q \left(x^2+z^2\right) x^2yz\,dV = \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} \left(x^2+z^2\right) x^2yz\,dz\,dy\,dx = \frac{684}{35} \approx 19.543,$$

$$I_z = \iiint_Q \left(x^2+y^2\right) x^2yz\,dV = \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} \left(x^2+y^2\right) x^2yz\,dz\,dy\,dx = \frac{729}{35} \approx 20.829.$$

Thus, the moments of inertia of the tetrahedron Q about the yz-plane, the xz-plane, and the xy-plane are $117/35$, $684/35$, and $729/35$, respectively.

THANK YOU